METRIC

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PERFORMANCE SPECIFICATION

LUBRICATING OIL, AIRCRAFT TURBINE ENGINE, SYNTHETIC BASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

This specification covers the requirements for two grades of aircraft turbine engine lubricating oil.

1.2 Classification.

The lubricating oil will be furnished in the following grades, as specified:

Grade	NATO Code
3	O-148
4	O-163

Grade 3 is normally used in most applications and will be supplied when requisitioned, unless otherwise specified.

Grade 4 should be specified when higher viscosity or greater thermal stability is needed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB OH 45433-7017 or e-mailed to ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARD

FED-STD-791	Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-8188	Corrosion-Preventive, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-152, O-154, O-156, and O-167
MIL-DTL-85694	Spectrometric Oil Standards

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-290	Packaging and Marking of Petroleum Related Products

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE TECHNICAL ORDERS

TO 33-1-37-2	Joint Oil Analysis Program Manual Volume II, Spectrometric and Physical Test Laboratory Operating Requirements and Procedures

(Copies of this document are available online at https://www.tinker.af.mil/Home/Technical-Orders.)

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DEFENSE STANDARDIZATION PROGRAM OFFICE

SD-6	Provisions Governing Qualification

(Copies of this document are available online at https://quicksearch.dla.mil/.)

DEPARTMENT OF LABOR

29 CFR 1910.1200	Occupational Safety and Health Standards – Hazard Communication

(Copies of this document are available online at https://www.ecfr.gov.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org/.)

ASTM INTERNATIONAL

ASTM B152/B152M	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM D92	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D445	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D972	Standard Test Method for Evaporation Loss of Lubricating Greases and Oils
ASTM D2273	Standard Test Method for Trace Sediment in Lubricating Oils
ASTM D2532	Standard Test Method for Viscosity and Viscosity Change After Standing at Low Temperature of Aircraft Turbine Lubricants
ASTM D2603	Standard Test Method for Sonic Shear Stability of Polymer-Containing Oils
ASTM D4057	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D4177	Standard Practice for Automatic Sampling of Petroleum and Petroleum Products
ASTM D4636	Standard Test Method for Corrosiveness and Oxidation Stability of Hydraulic Oils, Aircraft Turbine Engine Lubricants, and Other Highly Refined Oils

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ASTM D5185	Standard Test Method for Multielement Determination of Used and Unused Lubricating Oils and Base Oils by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES)
ASTM D6595	Standard Test Method for Determination of Wear Metals and Contaminants in Used Lubricating Oils or Used Hydraulic Fluids by Rotating Disc Electrode Atomic Emission Spectrometry
ASTM E1	Standard Specification for ASTM Liquid-in-Glass Thermometers

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE AMS3217	Standard Elastomer Stocks Test Slabs
SAE AMS3217/1	Test Slabs, Acrylonitrile Butadiene (NBR-H) Medium-High Acrylonitrile, 65 – 75
SAE AMS3217/4	Test Slabs, Fluoroelastomer (FKM) 65 – 75
SAE AMS3217/5	Test Slabs, Fluorosilicone (FVMQ) 55 – 65
SAE AMS4616	Silicon Bronze Bars, Rods, Forgings, and Tubing 92Cu – 3.2Si – 2.8Zn – 1.5Fe Stress Relieved
SAE AMS5544	Nickel Alloy, Corrosion and Heat-Resistant, Sheet, Strip, and Plate 57Ni – 19.5Cr – 13.5Co – 4.2Mo – 3.0Ti – 1.4Al – 0.05Zr – 0.006B Consumable Electrode or Vacuum Induction Melted, Annealed
SAE ARP5088	Test Method for the Determination of Total Acidity in Polyol Ester and Diester Gas Turbine Lubricants by Automatic Potentiometric Titration

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence.
Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification.
The lubricating oils furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials.
The composition of this lubricating oil is not limited: however, organometallic compounds of titanium and silicone antifoam agents and known or suspected human carcinogens (as defined

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by the Occupational Safety and Health Standards – Hazard Communication, 29 CFR 1910.1200) are prohibited. If a tricresyl phosphate additive is used, the tricresyl phosphate additive shall not contain more than 0.2 percent by weight ortho cresol containing isomers tricresyl phosphate. The engine lubricating oil shall have no adverse effect on the health of personnel when used for its intended purpose. Recycled base stocks are permitted; however, each batch must be fully tested in accordance with all qualification requirements of this specification. The manufacturer may be required to submit certification of conformance to 3.2 (see 6.2).

3.3 Chemical and physical requirements.

All classifications of the finished lubricating oil shall conform to the requirements listed in section 3, Tables I, II, and III when tested in accordance with the applicable test methods.

TABLE I. Chemical, physical, and performance requirements.

Characteristic	Requirement		Test Method	Reference Paragraph
	Grade 3	Grade 4		
Particulate contamination (3 micron filter)			FED-STD-791 FTM 3013	
Contaminant, mg/liter	5.0 max	5.0 max		
Filtering time, min				
0.95 liter (1 quart)	30 max	30 max		
223 milliliter (7.5 ounce)	7.0 max			
Total acid number (TAN), mg KOH/g	0.40 max	0.50 max	SAE ARP5088	
Viscosity at 205 °C, mm^2/s (cSt)		1.1 min	ASTM D445	
Viscosity at 100 °C, mm^2/s (cSt)	3.0 min	4.0 min	ASTM D445	
Viscosity at 40 °C, mm^2/s (cSt)	11.5 min	17.0 min	ASTM D445	
Viscosity at -51 °C, mm^2/s (cSt)				
at 35 minutes	13×10^3 max	20×10^3 max	ASTM D2532	
at 3 hours	13×10^3 max	20×10^3 max	ASTM D2532	4.4.1
at 72 hours	13×10^3 max	20×10^3 max	ASTM D2532	
Flash point, °C	210 min	210 min	ASTM D92	
Evaporation loss at 204 °C, %	30 max	15 max	ASTM D972	4.4.2
Foaming (static)			FED-STD-791	4.4.3
Foam volume, mL	100 max	100 max	FTM 3213	
Foam collapse time, sec	60 max	60 max		

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TABLE I. Chemical, physical, and performance requirements – Continued.

Characteristic	Requirement		Test Method	Reference Paragraph
	Grade 3	Grade 4		
Deposition test (WADC)			FED-STD-791 FTM 5003	
Deposit rating	1.5 max	0.7 max		
Acid number change, mg KOH/g	20 max	20 max		
Viscosity at 40 °C, % change	100 max	100 max		
Oil consumption, mL	100 max	100 max		
Lead corrosion, g/m^2	9.3 max	9.3 max	FED-STD-791 FTM 5321	4.4.4
Corrosiveness test at 232 °C			FED-STD-791 FTM 5305	4.4.5
Silver, g/m^2	±4.5 max	±4.5 max		
Bronze, g/m^2	±4.5 max	±4.5 max		
Thermal stability and corrosivity at 274 °C			FED-STD-791 FTM 3411	4.4.6
Viscosity change, %		5.0max		
Total acid number change, mg KOH/g		6.0 max		
Metal weight change, mg/cm^2		4.0 max		
Shear stability, viscosity loss %		4 max	ASTM D2603	4.4.7
Trace metal content by RDE-AES, ppm (mg/kg), maximum			ASTM D6595	4.4.8
Aluminum (Al)	2	2		
Iron (Fe)	2	2		
Chromium (Cr)	2	2		
Silver (Ag)	1	1		
Copper (Cu)	1	1		
Tin (Sn)	11	11		
Magnesium (Mg)	2	2		
Nickel (Ni)	2	2		
Titanium (Ti)	1	1		
Silicon (Si)	2	2		
Zinc (Zn)	2	2		
Lead (Pb)	2	2		
Molybdenum (Mo)	3	3		
Boron (B)	2	2		

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TABLE I. Chemical, physical, and performance requirements – Continued.

Characteristic	Requirement		Test Method	Reference Paragraph
	Grade 3	Grade 4		
Trace metal content ICP-AES, ppm (mg/kg), maximum			ASTM D5185	4.4.8
Aluminum (Al)	2	2		
Iron (Fe)	2	2		
Chromium (Cr)	2	2		
Silver (Ag)	1	1		
Copper (Cu)	1	1		
Tin (Sn)	4	4		
Magnesium (Mg)	2	2		
Nickel (Ni)	2	2		
Titanium (Ti)	1	1		
Silicon (Si)	2	2		
Zinc (Zn)	2	2		
Lead (Pb)	2	2		
Molybdenum (Mo)	3	3		
Boron (B)	2	2		
Compatibility			FED-STD-791 FTM 3403	4.4.9
Turbidity	None	None		
Sediment, mL/200 mL oil, maximum				
MIL-PRF-7808	0.005	0.005		
MIL-PRF-23699	0.005	0.005		
MIL-PRF-8188	0.05	0.05		
Storage stability				
Low temperature, 6 weeks at -18 °C	No crystallization, separation or gelling			4.4.10
Extended, after 3 years, ambient temperature	Meet conformance inspection (see 4.3)			4.4.11

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TABLE II. Chemical, physical, and performance requirements: elastomer compatibility.

Standard Elastomer Stock	FED-STD-791 Method	Test Time at Temp (Hour/°C)	Swell Percent Min-Max	Tensile Strength, Percent Change, Max	Elongation Percent Change, Max	Hardness Change, Shore A, Max
SAE AMS3217/1	3604	168/70	12-35			
SAE AMS3217/4	3432	72/175	2-25	50	50	20
SAE AMS3217/5	3432	72/150	2-25	50	50	20

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TABLE III. Chemical, physical, and performance requirements: Corrosion and oxidation stability.

Test Conditions Using ASTM D4636 Reference Paragraph 4.4.12		Post Test Oil Properties				Post Test Metal Specimen Weight Change, mg/cm², max						
Duration	Temperature	Change in Viscosity at 40 °C %	Change in Neutralization Number, mg KOH/g	Sludge Volume, ml	Weight Loss, %	Al	Ag	Bz[6/]	Fe	M-50	Mg	Ti
Grade 3:												
96 hours	175 °C[2/ 3/]	-5 to +15	2.0 max	no visible sludge[5/]	4.0 max	±0.2	±0.2	±0.4	±0.2	±0.2	±0.4	±0.2
96 hours	200 °C[2/ 4/]	-5 to +25	4.0 max		4.0 max	±0.2	±0.2	±0.4	±0.2	±0.2	±0.4	±0.2
96 hours	200 °C[1/ 3/]	-5 to +25	4.0 max		4.0 max	±0.2	±0.2	±0.4	±0.2	±0.2	±0.4	±0.2
Grade 4:												
96 hours	200 °C[2/ 3/]	-5 to +18	2.0 max	no visible sludge[5/]	4.0 max	±0.2	±0.2	±0.4	±0.2	±0.2	±0.4	±0.2
40 hours	220 °C[2/ 4/]	-5 to +25	4.0 max		4.0 max	±0.2	±0.2	±0.4	±0.2	±0.2	±0.4	±0.2
40 hours	220 °C[1/ 3/]	-5 to +25	4.0 max		4.0 max	±0.2	±0.2	±0.4	±0.2	±0.2	±0.4	±0.2

NOTES:

1/ Standard test procedure using intermediate sampling is required.

2/ Alternate procedure 1 without intermediate sampling is required.

3/ Qualification test only.

4/ Qualification and conformance test.

5/ If sludge is visible, it shall be determined not to exceed 0.2% max.

6/ Bronze

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3.4 Bench performance requirements.
Bench performance requirements shall be as specified in Table IV.

TABLE IV. Bench performance requirements.

Characteristic	Requirement		Test Method	Reference Paragraph
	Grade 3	Grade 4		
Bearing Deposition:			FED-STD-791 FTM 3410	4.4.13
Deposit Rating, max	60	45		
Filter Deposit Weight, g (max)	2.0	1.0		
Oil Consumption, mL (max)	1440	960		
Viscosity at 40 °C, % change	-5 to +25	-5 to +15		
Acid Number Change, mg KOH/g (max)	1.0	1.0		
Metal Specimen Weight Change, mg/cm² (max)	±0.2	±0.2		
Gear Load Carrying Capacity: Relative Rating as % Herco A – average of six determinations, min.	85%		FED-STD-791 FTM 6508	4.4.14

3.5 Full-scale performance requirements.

3.5.1 Engine endurance.
The oils shall be tested in a turboshaft engine in accordance with 4.4.15 to evaluate acceptability for turbine engine use. The posttest condition of the engine shall indicate no excessive deposits, wear, corrosion, or adverse conditions attributable to the test oil.

3.6 Workmanship.
The finished lubricating oil shall be transparent and uniform in appearance, and free from cloudiness, suspended matter, or other adulterations when examined visually by transmitted light.

4. VERIFICATION

4.1 Classification of inspections.
The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 Qualification inspection.

Qualification inspection shall consist of testing to all the requirements specified in section 3 including Tables I, II, III, and IV. When required by the qualifying activity (see 6.3), additional evaluations (engine and flight test service evaluation) may be required on candidate formulations that were the result of unique or unusual formulation or manufacturing technologies. The general outline of the qualification process (see 6.3) is described in the SD-6.

4.2.1 Qualification inspection sample.

The qualification test sample shall consist of 380 liters (100 gallons) of the finished lubricating oil packaged in 208-liter (55 gallon) drum containers. In addition, 19 liters (5 gallons) of the base oil blend (without additives), 0.95 liter (1 quart) of each base stock blend ingredient (without additives), and a minimum of 100 grams (3.5 ounces) of each additive ingredient used in the manufacture of the qualification test sample shall be submitted prior to qualification. Each lubricant ingredient shall be from the same bulk lot (see 6.4) used in preparation of the qualification test sample.

4.2.2 Requalification.

Requalification shall be required when any change is made on source of manufacture, purity, composition of the lubricating oil base stocks or additives. A minor change in the oil formulation may be made without requalification testing, but only after notification to, and approval by, the qualifying activity (see 6.3).

4.3 Conformance inspection.

Conformance inspection of production lots shall consist of all the tests specified in Table V. Failure of production lots to pass any of the conformance tests shall be cause for rejection of the lot.

4.3.1 Sampling and inspection.

Each bulk lot (see 6.4) of material shall be sampled in accordance with ASTM D4057 or ASTM D4177 for the conformance inspection tests in Table V. Inspections shall meet or exceed requirements specified in the government contract or order.

4.3.2 Examination of filled containers.

A random sample of filled containers from each packaged lot (see 6.4), taken in accordance with ASQ/ANSI Z1.4, shall be examined for conformance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and markings. Reject any container having one or more defects or for being under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of ASQ/ANSI Z1.4, reject the lot represented by the sample.

4.3.3 Conformance inspection test report.

A copy of the conformance inspection report on each lot of oil produced for U.S. Government use shall be provided via electronic means or mailed to the Aerospace Systems Directorate, AFRL/RQTM, Bldg. 490, 1790 Loop Road North, Wright-Patterson AFB OH 45433-7103.

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TABLE V. Conformance tests.

Characteristic	Requirements	
	Reference Table	Additional Paragraphs
Workmanship		3.6
Corrosion and oxidation stability: Alt procedure 1, w/o intermediate sampling Grade 3: 96 hours at 200 °C Grade 4: 40 hours at 220 °C	 III III	
Trace metal content, ppm (mg/kg), max	I	4.4.16.1
Flash point, °C	I	
Evaporation loss, at 204 °C, %	I	
Foaming (static), foam volume, ml, max	I	
Deposition, WADC	I	
Corrosiveness test at 232 °C	I	
Thermal stability and corrosivity at 274 °C	I	
Particulate contamination, mg/liter	I	4.4.16.2
Total acid number (TAN), mg KOH/g	I	
Viscosity, mm^2/s (cSt) at 100 °C at 40 °C at -51 °C, 35 minutes, 3 hours	 I	 4.4.16.3
Gear load carrying capacity	IV	4.4.16.4

4.4 Test methods.
All tests shall be performed in accordance with Tables I, II, III, IV, and V.

4.4.1 Low temperature viscosity.
Results at the 3-hour determination shall not differ from the 35-minute determination by more than 6.0 percent.

4.4.2 Evaporation loss.
Use a 6.5-hour test period and a bath temperature of 204 ±1 °C. An ASTM E1 thermometer No. 80F or digital equivalent shall be used. Air temperature shall be maintained at 204 ±1 °C, using a pre-heater if necessary.

4.4.3 Foaming.
Foam collapse time shall be defined as the time required for the foam to collapse to no more than a single ring of bubbles at the lubricant surface periphery around the cylinder wall and air inlet tube. If the ring of bubbles around the cylinder wall contains segments having two or more layers of bubbles and the difference in height of the foam in the ring is not greater than 10 mL, complete foam collapse is the point at which a break occurs in the ring of bubbles without subsequent reforming of the ring.

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4.4.4 Lead corrosion.
Multiply mg/in² by 1.55 to convert to g/m².

4.4.5 Corrosiveness test at 232 °C.
Silicon Bronze (see SAE AMS4616) shall be used in lieu of copper (see ASTM B152/B152M). In order to minimize vapors, Erlenmeyer flasks with proper ground joints (24/40) may be used with elbow adapters and tubing, capturing vapors in a liquid trap. Multiply mg/in² by 1.55 to convert to g/m².

4.4.6 Thermal stability and corrosivity.
Posttest oil viscosity sample shall be compared with viscosity of new oil samples tested at 40 °C. Posttest oil total acid number is compared to the total acid number of the new oil sample. The viscosity shall be determined in accordance with ASTM D445. The total acid number shall be determined in accordance with SAE ARP5088.

4.4.7 Shear stability.
Use an irradiation period of 30 minutes on a 30 mL oil sample at a power setting which causes 11.5 ±0.5 percent viscosity loss to a 30 mL sample of ASTM Reference Fluid A, when irradiated five minutes. ASTM Reference Fluid A is a petroleum oil containing a polymer capable of being broken down by turbulence at high rates of shear, typical viscosities are 10.8 mm²/s (cSt) at 100 °C and 58 mm²/s (cSt) at 40 °C. ASTM Reference Fluid A may be obtained by vendors listed in the ASTM D2603 method.

4.4.8 Trace metal content.
The trace metal content of the oil for qualification testing shall be determined both with a Rotating Disc Electrode - Atomic Emission Spectrometer (RDE-AES) (see ASTM D6595) and Inductively Coupled Plasma – Atomic Emission Spectrometer (ICP-AES) (see ASTM D5185).

The RDE-AES shall be standardized in accordance with Air Force TO 33-1-37-2 using MIL-DTL-85694 spectrometric standards. Immediately after standardizing the spectrometer, five determinations of the oil for trace metal content shall be performed. The average of the five determinations shall be reported. Samples which require trace metal content determinations by RDE-AES contact the qualification activity (see 6.3) with a request for analysis.

In accordance with ASTM D5185, modifications that allow for the adjustments in the dilution of the sample are allowed to accurately measure 2 ppm (mg/kg) of metals in the ICP-AES.

4.4.9 Compatibility.
The compatibility test shall be performed in accordance with FED-STD-791, Method 3403.2. The lubricating oil shall be mixable with selected referee lubricating oils qualified under this specification, MIL-PRF-8188, and MIL-PRF-23699, without turbidity. Upon completion of the 168-hour oven period, the test flasks shall be stored in the dark, at a room temperature of 25 ±5 °C, for 21 days before visual inspection for turbidity and centrifuging. Centrifuge tubes defined in ASTM D2273 with a small scale division of 0.005 mL or less shall be used. The volume of sediment shall be determined in accordance with ASTM D2273. The sediment shall be centrifuged from a 200 mL oil sample without the use of solvent. The volume of sediment for mixtures which involve oils covered under this specification and MIL-PRF-23699 shall be less than 0.005 mL per 200 mL of oil. The volume of sediment for mixtures that involves oils covered under MIL-PRF-8188 shall be less than 0.05 mL of oil. If the amount of sediment collected after the exposure period is greater than the limit specified in Table I, additional testing may be

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performed on the mixture to determine that its performance meets the requirements of this specification. The additional testing may include all of the tests specified in this specification.

4.4.10 Low temperature storage.
Three 946 mL (1-quart) samples of oil shall be stored in a cold chamber maintained at -18 ±2.5 °C for six weeks. At the end of the storage period, the oil shall be visually inspected for evidence of crystallization, additive separation, and gelling.

4.4.11 Extended storage.
Forty-five liters (12 gallons) of the lubricating oil packaged in 946 mL (1-quart) containers from the first production run shall be stored at an ambient temperature not lower than -40 °C and not greater than 60 °C for periods up to three years. After storage for the specified time, the lubricating oil shall be examined for conformance to the inspection requirements of conformance inspection (see 4.3).

Tentative qualification approval will be given to products meeting qualification inspection (see 4.2), pending successful completion of the extended storage stability test. Failure to pass this test shall be cause for withdrawal of approval.

4.4.12 Corrosion and oxidation stability.
The corrosion and oxidation stability test shall be performed in accordance with ASTM D4636 with the following modifications:

a. If no visible sludge is present, it is not required to perform the Sludge Determination as described in ASTM D4636.
b. Duplicate tests shall be run at all test temperatures. The individual results shall be reported for qualification testing.
c. Intermediate and posttest oil viscosity samples are compared with viscosity of new oil samples tested at 40 °C. Intermediate and posttest oil total acid number are compared to the total acid number of the new oil sample in accordance with SAE ARP5088.

4.4.13 Bearing deposition.
To determine compliance with the requirements of Table IV, bearing test determinations shall be conducted in accordance with FED-STD-791, Method 3410, severity Type 3. Metal specimens (washer-shaped) shall be the same as in ASTM D4636 except Waspaloy (see SAE AMS5544) is substituted for magnesium.

4.4.14 Gear load carrying capacity.
The average of six determinations shall not be less than 85 percent of the reference oil (Hercolube A) when tested in accordance with FED-STD-791, Method 6508 and Table IV. All six determinations shall be made using the same machine. The reference oil average rating used to obtain the relative ratings shall also be reported. Only the Ryder gear machines having a reference oil average rating of 2,100-2,600 lbs/in after eight determinations are acceptable.

4.4.15 Engine endurance test.
The oil shall be subjected to an accelerated endurance test in a turboshaft engine for a period sufficient to determine its performance characteristics, as specified by the qualifying activity (see 6.3). Periodic oil samples shall be withdrawn and tested during, and upon completion of, the test. The condition of the periodically sampled used oil shall indicate no excessive changes in

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performance characteristics. A posttest condition of the engine shall indicate no excessive deposits, wear, corrosion, or other adverse conditions attributable to the test oil.

4.4.16 Conformance inspection testing methods.
The following information is in support of the conformance testing required in Table V.

4.4.16.1 Trace metal content.
Trace metal content for conformance testing can be tested using either RDE-AES or ICP-AES.

4.4.16.2 Particulate contamination.
Recording of filtering time is not required for conformance testing.

4.4.16.3 Viscosity at -51 °C.
The initial viscosity shall be determined at 35 minutes after the viscometer is placed in the bath and again at three hours after completion of the initial viscosity. These results shall not differ by more than 6.0 percent.

4.4.16.4 Gear load-carrying capacity test.
Test shall be performed on the first production lot of each qualified lubricating oil supplied to the procuring activity.

5. PACKAGING

5.1 Packaging.
For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.
The lubricating oil is intended for use in specific models of aircraft turbine engines, helicopter transmissions, accessory and auxiliary power units, and other types of equipment that require a synthetic base oil. Grade 3 is designed for operation within the approximate bulk oil temperature range of -51 °C to 177 °C and is intended for normal use unless otherwise specified. Grade 4 is intended for aerospace applications, which require higher viscosity and greater thermal stability for hotter running engine designs.

6.2 Acquisition requirements.
Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Grade including any amendments (if not specified, Grade 3 will be issued).

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c. Type and size of containers required.

d. Quantity desired.

e. If certification of conformance to material prohibitions is required (see 3.2).

f. Submittal of conformance test results (see 4.3.3).

g. Packaging requirements (see 5.1).

6.3 Qualification.

With respect to products requiring qualification, awards will be made only for products which are, at the time of the award of contract, qualified for inclusion in Qualified Products List QPL No. 7808 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Aerospace Systems Directorate, AFRL/RQTM, Bldg. 490, 1790 Loop Road North, Wright-Patterson AFB OH 45433-7103. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil/.

6.3.1 Requesting qualification.

At the initiation of the qualification process, prospective suppliers must forward a written request for such action to the Aerospace Systems Directorate, AFRL/RQTM, Bldg. 490, 1790 Loop Road North, Wright-Patterson AFB OH 45433-7103. The letter must contain the following:

a. Request for authorization to submit test sample for qualification.
b. Certified test report that contains data on the specific batch of test sample to be submitted showing results of the tests specified herein except compatibility, low temperature storage, extended storage stability, bearing deposit test, gear load carrying ability, and engine endurance.
c. Complete formulation data, including chemical composition, percentages of each ingredient, the manufacturer and trade name of each ingredient, and the purity of each ingredient. Formulation data will be respected as highly proprietary information. The manufacturer must also submit the SDS of the candidate product and for each of the additive components used in the formulation.
d. Verification that the composition of the test sample complies with the requirements of 3.2.
e. Identification of the manufacturing site of the specific batch test sample to be submitted.

6.3.2 Qualification sample.

AFRL/RQTM will respond in writing with a formal "letter of authorization" providing detailed instructions for the submission of product samples. Upon receiving authorization from AFRL/RQTM, these samples will be forwarded to: Aerospace Systems Directorate, AFRL/RQTM, Bldg. 490, 1790 Loop Road North, Wright-Patterson AFB OH 45433-7103. Each sample must be plainly identified by a securely attached, durable label marked with the following information:

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**QUALIFICATION INSPECTION SAMPLE LUBRICATING OIL,
AIRCRAFT TURBINE ENGINE, MIL-PRF-7808**

Type of sample: (base stock, additive, or finished oil)

Classification of oil: (Grade 3 or Grade 4)

Name of manufacturer:

Product code number:

Batch number:

Date of manufacture:

Submitted by (*NAME)* on *(DATE)* for qualification inspection in accordance with MIL-PRF-7808 under authorization of (*reference authorization letter, see 6.3.2)*.

6.3.3 QPL recertification.
Items qualified to MIL-PRF-7808 require recertification every two years even if an oil formulation does not have any changes. The most recent production batch data for each qualified product should be sent to the qualifying activity (see 6.3).

6.3.4 Heat transfer properties.
There is no qualification requirement listed in section 3 for specific heat or thermal conductivity of potential products. The heat transfer properties of a given chemical class, such as polyol ester lubricants, are very consistent. Therefore, such a requirement is not warranted.

6.4 Definitions.

Bulk lot	A bulk lot is defined as an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single, isolated container or manufactured by a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in ingredient material.
Packaged lot	A packaged lot is defined as an indefinite number of 208-liter (55 gallon) drums or smaller unit packages of identical size and type offered for acceptance and filled with a homogeneous mixture of material from one isolated container or filled with a homogeneous mixture of material manufactured by a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in ingredient material.
Reblend	A reblend lubricating oil is an original qualified lubricating oil as specified in 4.2, in which one or more ingredients have been blended by a manufacturer other than the manufacturer of the original formulation. A bulk lot of the reblended oil will be subjected to the qualification tests (see 4.2). The engine performance requirements (see 3.5.1) may be waived, at the discretion of the qualifying activity, if other tests demonstrate

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equivalence to the original formulation. Reblend approvals may be initiated by the process described in 6.3.

Rebrand — A rebrand lubricating oil is a qualified, fully-formulated oil which successfully passed all qualification tests (see 4.2) and is manufactured by the original formulator at the original manufacturing site but which is packaged by a supplier other than the manufacturer of the fully-formulated oil. Rebrand approvals may be initiated by the process described in 6.3.

6.5 Shelf-life.

This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume I, Shelf-Life Management Program: Program Administration; and Volume 2, Shelf-Life Management Program: Materiel Quality Control Storage Standards. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.6 Subject term (key word) listing.

Corrosion inhibition
Ester lubricant
Thermal stability

6.7 International standardization agreement implementation.

This specification implements Air Force Interoperability Council (AFIC) Air Standard 4024, Interchangeability Chart of Standardised Aviation Fuels, Lubricants and Associated Products; and STANAG/AFLP-1135, Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST Online database at https://assist.dla.mil.

6.8 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR4
Navy - AS
Air Force - 20

Review activities:
Army - AV
Navy - SH
Air Force - 68
DLA - PS

Preparing activity:
Air Force - 20

Agent:
Air Force - 110

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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